

Problem Solving and Calculations 12

Set 12: Atoms and X-rays

Notes

- What is the difference between an atom in the ground-state and an atom in the excited state?
 - How do atoms become excited?
- Explain why an iron bar at 200 °C appears grey, at 800 °C appears red, and at 1400 °C appears white.
- Bohr developed a formula for calculating the energies of the energy levels of the hydrogen atom.

$$E_n = -\frac{13.6 \text{ eV}}{n^2}$$

where:

E_n = is the energy of an electron at level n
 $n = 1, 2, 3$, etc

Use this formula to find the values of the first five energy levels of the hydrogen atom, in eV.

- An energy level diagram for the hydrogen atom is shown.
 - For each of the downward transitions listed below, determine:
 - the energy of the photon the electron emits,
 - the frequency of the photon the electron emits,
 - the type of electromagnetic radiation to which the photon belongs
 - $E_2 \rightarrow E_1$
 - $E_4 \rightarrow E_2$
 - $E_3 \rightarrow E_2$
 - $E_5 \rightarrow E_3$
 - A violet line in the visible line emission spectrum of hydrogen gas has a wavelength of 434 nm. Between which two levels must an electron fall to emit this photon?
- Electrons with energy of 45 keV bombard a metal target in an X-ray tube. The energies of some of the scattered electrons are 15 keV and 5 keV.
 - Determine the highest energy X-ray that this X-ray tube could produce.
 - Explain the X-ray's origin.
 - Find the energy of the X-ray photon produced by each of the two scattered electrons.
 - Would you classify these X-rays as *hard* or *soft*? Justify your answer.
 - The X-ray spectrum the tube produces also shows a significant X-ray peak at 1.0 nm. These X-rays come from electrons dropping into an inner energy level of a target metal atom. Determine the difference between the energies of the two levels.
- Many minerals that have dull colours in daylight fluoresce under ultraviolet light. Explain how fluorescence occurs, using a simple energy level diagram.

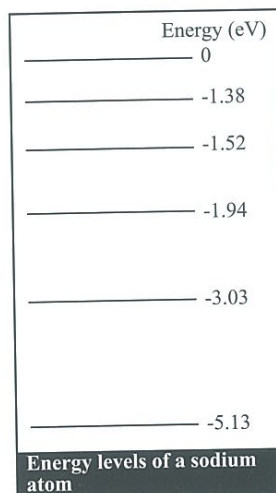
_____	$E_5 = -0.87 \times 10^{-19} \text{ J}$
_____	$E_4 = -1.36 \times 10^{-19} \text{ J}$
_____	$E_3 = -2.41 \times 10^{-19} \text{ J}$
_____	$E_2 = -5.43 \times 10^{-19} \text{ J}$
_____	$E_1 = -21.8 \times 10^{-19} \text{ J}$
Energy levels	

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7. Some cities, such as Adelaide and Perth, use sodium vapour lamps for lighting major streets. Perth also uses sodium vapour lamps for cross-walk lighting. Our eyes are particularly responsive to the bright yellow light of sodium vapour lamps. The line emission spectrum of sodium shows a yellow line and no other colours. The diagram shows some of the energy levels of the sodium atom. The diagram is not drawn to scale.



- What energy will eject an electron from a sodium atom if the electron is in the lowest energy level?
 - The characteristic yellow colour of the sodium lamp is due to the emission of photons of wavelength 589 nm. Between which two levels do electrons fall to emit this colour?
 - The line emission spectrum of a low pressure sodium vapour lamp shows that the yellow line is really two very closely-spaced fine yellow lines of wavelengths 589.0 nm and 589.6 nm. This happens because the upper energy level really consists of two very closely spaced energy levels. Determine the energy difference between these two levels.
 - All other electron transitions within the sodium atom are either in the infrared region or ultraviolet region. Sketch an energy level diagram showing two electron transitions producing infrared radiation and two electron transitions producing ultraviolet radiation.
- 8.
- Describe with the aid of a diagram, the construction and principle of an X-ray tube.
 - What is the thermionic effect and how is this used in an X-ray tube?
 - Why is the anode usually made to rotate when an X-ray tube is operating?
 - The anode normally consists of a tungsten target embedded in copper. Why are these two materials chosen?
 - Sketch a graph of the intensity of X-rays produced against their frequency.
 - Discuss each of the significant features of the graph and the physical principles that give rise to these features.
 - How would the graph change if:
 - A different material was used for the target
 - The current through the heated filament was increased
 - The accelerating potential was increased?
 - Why do X-rays have a maximum frequency for an given accelerating voltage?

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9. X-rays were once used to check the fit of shoes. Why has this practice been discontinued?



Notes

10. Varun remembered that his Physics teacher said that in an X-ray machine about 99% of the energy of the incident electrons was converted to heat. When electrons are accelerated through 100 kV, Varun's interpretation was that the maximum energy of the X-ray photon produced was 1% of the 100 keV. Emily however, claims this is not true and that photons of maximum energy 100 keV were produced. Who is right and why?
11. X-rays are created by accelerating electrons from a cathode across a potential difference of 60 000 V. Find:
- The energy of each electron
 - The final speed of each electron
 - The minimum wavelength of the X-ray photons created by the collision of the electrons with the target.
12. What potential difference is needed to accelerate electrons in order that X-rays of wavelength 10^{-10} m are produced?
13. What is the shortest-wavelength X-rays emitted by electrons striking the face of a 33.5 kV TV picture tube? Why can we not calculate the longest wavelengths?
14. In an X-ray tube, the high voltage between the filament and the target is V. After being accelerated through this voltage, an electron strikes the target where it is decelerated (by positively charged nuclei) and in the process one or more X-ray photons are emitted.
- Show that the highest-energy photon will have a wavelength given by:
$$\lambda_0 = \frac{nc}{eV}$$
 - Determine the shortest wavelength and highest frequency of the X-rays emitted when accelerated electrons strike the face of a 30 kV television tube.
15. The Standard Model postulates the existence of several quark types. Protons and neutrons are made of combinations of three 'up quarks' and 'down quarks'. An up quark has an electric charge of $+\frac{2}{3}e$ and a down quark has a charge of $-\frac{1}{3}e$. Predict the quark combination that makes up:
- a proton
 - a neutron

- 14 a) 383 mm from the balcony
 15 $1.86 \times 10^5 \text{ N}$; $1.95 \times 10^5 \text{ N}$
 20 7.76 kg; 2.57 m from the T_1 end.
 21 60 cm
 23 a) 137 N
 b) lower hinge 172 N toward the door at 36.9° above the horizontal
 upper hinge 172 N toward the wall at 36.9° above the horizontal
 24 a) $1.95 \times 10^3 \text{ N}$
 b) vertical = 109 N upward, horizontal = $1.83 \times 10^3 \text{ N}$ out from wall
 c) $1.84 \times 10^3 \text{ N}$ out from wall at 3.40° above the horizontal
 25 tension = $1.09 \times 10^3 \text{ N}$; force at hinge = 551 N out from the wall and 10.1° below the horizontal
 26 2.07 m from P
 27 a) 156 N
 b) vertical component = 147 N, horizontal component = 156 N
 c) 214 N to right and 43.3° below the horizontal
 28 a) $2.39 \times 10^5 \text{ N}$
 b) $2.97 \times 10^5 \text{ N}$
 29 a) 18.4 kg
 b) 144 N at 26.9° above the horizontal

Electricity and Magnetism

- Set 6 5 0.042 N m
 10 2.4 N
 12 0.075 N downward
 Set 7 1 0.57 V
 2 a) 2.3 A
 5 a) 0.21 mV
 5 b) $2.1 \times 10^{-4} \text{ Wb s}^{-1}$
 7 7.1 mV
 8 a) 39 mV
 8 d) 7.9 mA
 9 a) $8.0 \times 10^{-4} \text{ V}$
 11 225 turns
 12 a) 2.5 m s^{-1}
 12 b) $1.0 \times 10^{-4} \text{ N}$
 13 a) 83 mT
 Set 8 1 a) 60 kW
 1 b) 300 J
 1 c) 150Ω
 2 a) 14.6 m^2
 2 b) 1.25Ω
 3 a) 83 A
 3 b) 18Ω
 6 a) 0.0200 times
 6 b) 10 000 turns
 6 c) 0.0196 times
 7 a) 23Ω
 7 b) $1.3 \times 10^3 \text{ C}$
 8 a) 100 A
 10 a) 31.3 kW
 10 b) 0.31 kW
 12 a) 2.00 kW
 12 b) 0.125 m
 13 a) 6.25 kW
 15 35 km

Particles, Waves and Quanta

- Set 9
- 1 5 °C
 - 2 420 m
 - 3 82 m
 - 4 a) 50 mm
 - 4 b) 0.4 m s^{-1}
 - 4 c) 8.4 Hz
 - 5 a) 0.5 Hz
 - 5 b) 2 s
 - 6 a) 10 mm
 - 6 b) 8 ?s
 - 6 c) 125 kHz
 - 9 a) 1.3 m
 - 9 b) 17 mm
 - 9 c) 4.9 mm
 - 9 d) 3.4 km
 - 13 b) too short
 - 13 c) 0.29 s
 - 14 a) George
 - 14 b) Jane
 - 14 c) George
- Set 10
- 3 c) 34 Hz to 17 kHz
 - 5 a) 1.9 mm in air; 8.1 mm in water
 - 6 72.8 m
 - 7 a) 100
 - 7 b) 0.8°
 - 7 c) 10 scans
 - 12 708 Hz
 - 13 a) 512 m s^{-1}
 - 13 c) 84 Hz, 168 Hz, 262 Hz
 - 18 a) 0.64 m
 - 18 b) 328 m s^{-1}
 - 18 c) closed
 - 19 a) violin
 - 19 b) double bass
- Set 11
- 1 a) 3.0 GHz
 - 1 b) 10cm
 - 2 red: $4.41 \times 10^{14} \text{ Hz}$; $2.92 \times 10^{-19} \text{ J}$
orange: $5.17 \times 10^{14} \text{ Hz}$; $3.43 \times 10^{-19} \text{ J}$
green: $6.00 \times 10^{14} \text{ Hz}$; $3.98 \times 10^{-19} \text{ J}$
 - 3 0.001:1; 1000:1
 - 4 a) red
 - 4 b) $2.87 \times 10^{-19} \text{ J per photon}$
 - 4 c) $1.00 \times 10^5 \text{ W m}^{-2}$
 - 4 d) 100:1
 - 5 7.5×10^{25} photons per second
 - 6 5.1×10^{10} photons per second
 - 7 a) 420 m
 - 7 b) $4.3 \times 10^9 \text{ J}$
- Set 12
- 3 $E_1 = -13.6 \text{ eV}$
 $E_2 = -3.4 \text{ eV}$
 $E_3 = -1.5 \text{ eV}$
 $E_4 = -0.85 \text{ eV}$
 $E_5 = -0.54 \text{ eV}$

- 4 a) (i) 2.4×10^{15} Hz; UV
 (ii) 6.14×10^{14} Hz; visible
 (iii) 4.56×10^{14} Hz; visible
 (iv) 2.32×10^{14} Hz; IR
 4 b) 4.58×10^{-19} J; E_5 to E_2
 5 a) 45 keV
 5 c) 30 keV; 40 keV
 5 e) 2.0×10^{-16} J
 7 a) 5.13 eV
 7 b) 2.11 eV; E_2 to E_1
 7 c) 0.002 eV
 11 a) 60 keV (9.6×10^{-15} J)
 11 b) 1.45×10^8 m s⁻¹
 11 c) 2.07×10^{-11} m
 12 12.4 kV
 13 3.71×10^{-11} m
 14 b) 4.14×10^{-11} m; 7.24×10^{18} Hz

Motion and Forces in Electric and Magnetic Fields

- Set 13 1 4.5×10^6 N C⁻¹
 3 b) 2.94×10^7 m s⁻¹, deflected by 8.9°
 3 e) 2.9×10^{-16} J
 3 f) 6.0×10^4 V m⁻¹
 5 a) 3.3×10^{-7} J
 5 b) 66 V
 6 100 V m⁻¹
 7 a) 5.00 keV
 7 b) 8.00×10^{-16} J
 8 a) 10.0 keV
 8 b) 1.6×10^{-15} J
 9 a) 5.6×10^7 V m⁻¹
 9 b) 2.4×10^{-15} J
 10 b) 3.6×10^{-5} N
 10 c) 10 mm; 0.090 V
 13 3.91×10^5 m s⁻¹
 14 b) 2500 eV (4.00×10^{-16} J)
 14 c) 2.96×10^7 m s⁻¹
 14 e) same
 14 f) less
 15 a) 4000 eV; 3.75×10^7 m s⁻¹
 15 b) 2000 V m⁻¹; 3.20×10^{-16} N
 16 b) 7.02×10^{15} m s⁻²
 Set 14 7 1.58×10^{-26} kg to 1.62×10^{-26} kg
 8 a) 41.8 m
 8 b) 6.28×10^{-3} s
 9 b) 2.4×10^{-14} N
 10 b) 6.3×10^7 C kg⁻¹; 4.8×10^7 C kg⁻¹
 Set 15 3 a) 1.10×10^5 V m⁻¹
 3 b) Yes
 4 c) They accelerate in a straight line at 1.76×10^{14} m s⁻²
 5 a) 7.26×10^7 m s⁻¹
 5 b) 1.76×10^{-4} m
 5 c) 1.52×10^{-11} s

- 6 a) $7.19 \times 10^6 \text{ m s}^{-1}$
- 6 b) $2.29 \times 10^7 \text{ Hz}$; $4.38 \times 10^{-8} \text{ s}$
- 6 c) 270 kV
- 7 c) $1.00 \times 10^5 \text{ m s}^{-1}$
- 8 a) $3.75 \times 10^{13} \text{ m s}^{-2}$ at right angles to the field
- 8 c) $3.92 \times 10^5 \text{ N C}^{-1}$
- 9 a) $3.04 \times 10^{-6} \text{ m}$
- 9 b) 480 N C^{-1}
- 10 b) $2.87 \times 10^7 \text{ m s}^{-1}$; $5.75 \times 10^7 \text{ m s}^{-1}$